Shuai Ma

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Education

Seattle University Seattle, WA, United States
Albers School of Business & Economics
M.S. in Business Analytics, STEM-designated 2018–2020

- Relevant coursework: programming for business analytics, applied econometrics, statistical learning, forecasting, big-data analytics, data management, data visualization, and data architecture.
- Capstone: *Influential Factors on Residential Property Prices Using Data-Mining Methods*; analyzed Pierce County housing transactions using statistical and machine-learning methods.

University of Cincinnati Cincinnati, OH, United States
College of Design, Architecture, Art, and Planning
M.S. in Architecture, research-based program; research and thesis focused on urban development 2009–2011

- Research training: urban theory, research methods, pre-thesis research, social-science research for planners, land-use control, international development policy, and thesis-based scholarly writing.
- Thesis: *Impact of High-Speed Rail Station Development on Surrounding Urban Form: The Beijing–Tianjin Case*.

Hebei University of Technology Tianjin, China
Project 211 and Double First-Class university
B.Eng. in Urban Planning and Design 2003–2008

- Undergraduate planning foundation: planning studios, regional planning, urban design, urban economics, urban geography, transportation planning, urban ecology, planning code, and spatial information technology.
- NACES course-by-course evaluation: equivalent to a U.S. bachelor's degree in Urban and Regional Planning.

Research Interests

Housing and urban policy; place-based public investment; housing and land-market capitalization; causal inference and quasi-experimental urban policy evaluation; spatial spillovers, affordability, neighborhood sorting, and distributional effects; climate adaptation and flood mitigation; GIS, spatial econometrics, and urban data science.

Research Papers and Writing Samples

Hazard or Amenity? Flood Risk, Waterway Proximity, and Neighborhood Inequality in Housing Prices 2025–Present
Sole-authored PhD writing sample; empirical study of Pierce County, Washington

- Examines how flood risk and waterway proximity are jointly capitalized into housing prices, and how this capitalization varies across neighborhoods differing in income, school quality, and crime, using 47,763 single-family sales linked to parcel records, FEMA flood-risk layers, and neighborhood indicators.
- Estimates hedonic regression models with tract-clustered standard errors and conducts spatial analysis to distinguish flood-risk effects from waterfront-amenity value.
- Provides the empirical foundation for a broader research agenda on how place-based public investments reshape housing markets, generate spatial spillovers, and distribute benefits and costs across communities.

Strategic Management's Foundational Silence 2025–Present
First-author manuscript in revision, coauthored with Rubina Mahsud and Greg Prussia; developed from a 2020 Strategic Management Society conference proposal

- Built and cleaned a corpus of 3,202 Strategic Management Journal articles and applied classical natural language processing methods, local open-source large language models, and cloud-based models to analyze the field's dominant vocabulary.
- Develops the concept of "foundational silence" to argue that poverty, war, disease, hunger, homelessness, and related human concerns should be treated as central social contexts rather than peripheral externalities.

Conference Presentations

Strategic Management Society 2060: From Effectiveness to Impact, from Success to Significance October 2020
Strategic Management Society Annual Conference, Virtual
Mahsud, R., and Ma, S. Presented by Shuai Ma.

Selected Research and Analytical Projects

Empirical Analysis of Guangzhou Residential Rental Markets 2021

Sole-authored applied econometric project using scraped public rental-listing data

- Collected and cleaned approximately 3,000 publicly available online rental listings and modeled Guangzhou rent variation using structural housing attributes, rental type, floor information, and district-level location indicators.
- Applied OLS regression, stepwise variable selection, VIF/kappa multicollinearity diagnostics, PCA, and exploratory clustering to evaluate key rental-price determinants.
- Identified floor area, number of bathrooms, lease type, and central-district location as the dominant rent drivers, with the preferred model explaining roughly 60% of rent variation.

Serviced Apartment and Long-Term Rental Housing Sector Studies 2016; 2021

Sole-authored mixed-methods housing-market research using field visits, web-scraped data, R, and Python

- Conducted field visits to dozens of serviced-apartment projects and compiled public online listing data to compare rents, occupancy, tenant profiles, product layouts, amenities, operators, and location patterns.
- Analyzed 987 Guangzhou serviced-apartment listings using descriptive statistics, cross-tabulation, chi-square tests, and tree-based feature-importance models to evaluate brand, district, accessibility, and unit-level rent drivers.

Highest and Best Use (HBU) Optimization for Business-Park Development 2021

Initiated and led internal real estate analytics study using market research, regulatory constraints, and Python linear programming

- Surveyed 11 competing business-park projects in Foshan to parameterize 16 candidate product typologies, including multi-floor manufacturing buildings, detached warehouses, and independent offices.
- Formulated a Python PuLP optimization model to maximize gross development value under FAR, building-height, plot-coverage, and green-space-ratio constraints.
- Tested alternative product-mix scenarios to compare value maximization with market absorption, translating HBU analysis into a data-driven development strategy.

Comparative Policy Study of New Industrial Land (M0) in the Greater Bay Area 2021

Sole-authored study of land-use regulation, industrial redevelopment, and zoning-like reform

- Compared M0 land-use policies across Shenzhen, Guangzhou, Dongguan, Foshan, Huizhou, Zhongshan, and Zhuhai, examining industry eligibility, site-selection rules, FAR requirements, subdivision and transfer restrictions, land tenure, and performance monitoring.
- Synthesized national, provincial, and municipal policy documents to evaluate how regulatory design supports industrial upgrading while limiting speculative real-estate conversion of industrial land.

Sponge City and Low-Impact Development Technical Research and Application 2017

Research leader; led a two-person, three-month applied research effort on stormwater adaptation

- Synthesized sponge-city policies, Low Impact Development (LID) technologies, and multi-city case materials into technical guidance for runoff control, rainwater reuse, water-quality treatment, and urban flood-risk reduction.
- Applied the framework across development projects in four cities (Shanghai, Shenzhen, Kunshan, and Jinjiang), estimating annual runoff-control rates, rainwater-harvesting volumes, and pre/post-development runoff at parcel-to-city scale.
- Evaluated permeable paving, green roofs, bioswales, rain gardens, wetlands, detention facilities, and runoff-control scenarios using catchment analysis, water-balance calculations, and stormwater simulation outputs.

Jiangsu Provincial Low-Carbon Urban System Strategy 2011

Core project-team member, Arup; province-scale carbon-accounting and policy-scenario analysis

- Helped build a multi-sector accounting model integrating activity data across land use, buildings, industry, transportation, water, waste, and ecological space for a 13-city provincial urban system.
- Produced baseline and policy-scenario estimates for 2005–2030 using per-capita and per-GDP intensity indicators, translating spatial and administrative data into planning-relevant quantitative evidence.

Technical Skills

Econometric and Quantitative Methods

Applied econometrics; hedonic price modeling; OLS with clustered standard errors; model selection and multicollinearity diagnostics; statistical learning and forecasting; scenario and baseline modeling; linear programming and optimization

Spatial and Data Methods

GIS, spatial analysis, spatial data integration, geospatial data processing, data visualization, web-scraped and administrative data

Programming and Software

Python, R, SQL, GeoPandas, ArcGIS/QGIS, Tableau, Excel, PuLP

Professional Experience

Founder & Principal Researcher, Beamlinked Consulting Co., Ltd. 2023–Present

- Founded an independent research and advisory practice focused on housing and real estate markets, land-use policy, and urban development, with project engagements in Shandong, Guangzhou, Chongqing, and other regions across China.
- Conducted market, policy, and spatial analyses for housing, business-park, and urban-development projects, integrating administrative information, online market data, land-use controls, and GIS to assess market demand, development feasibility, and product positioning.
- Developed quantitative and spatial workflows using Python, R, GIS, and administrative and web-collected data to support research and advisory assignments.

Vice Director, Property Asset Management, Ping An Group 2020–2023

Planning, research, and investment analysis lead within a real estate business unit

- Led housing and real estate market research and investment analysis for a business unit, producing recurring analytical reports for senior management on housing markets, urban policy, infrastructure investment, and industrial-park planning.
- Initiated quantitative studies on product positioning and highest-and-best-use optimization, evaluating how regulatory and market constraints affected product mix and gross development value.
- Conducted feasibility, market, and investment analysis for five industrial-park development projects with a combined planned investment of approximately RMB 3.75 billion, two of which are now operational.

Director, Planning and Development, China Fortune Land Development 2017–2018

- Directed planning and design across two public-private-partnership new-town districts (~3,200 acres each), spanning land use, urban design, road and infrastructure networks, flood-control and water systems, residential neighborhoods, parks and public facilities, within development contexts involving land acquisition and resident resettlement.
- Coordinated large-scale development across public and private actors, integrating land-use regulation, infrastructure provision, and real estate development.
- Co-developed a group-wide development guideline adopted as a standard for new real estate projects.

Project Manager, AtkinsRéalis / Atkins 2013–2017

- Led planning and design teams across town and master planning, business and industrial parks, commercial complexes, and urban development projects.
- Directed applied research on environmental and performance-based analysis (microclimate, energy, and building-physics simulation), applying quantitative modeling to development and design decisions.
- Advised project teams on green-building and sustainability standards (LEED, BREEAM, WELL, and equivalents).

Urban Planner, Arup 2011–2013

- Contributed to urban regeneration and redevelopment projects, including the Shanghai Expo Urban Best Practices Area (UBPA) and Knowledge & Innovation Community (KIC) renewal, alongside waterfront, transit-oriented, and regional planning with spatial and GIS-based analysis.
- Contributed to the Jiangsu Provincial Low-Carbon Urban System Strategy, supporting carbon-audit modeling and province-scale scenario analysis.
- Supported a National Basic Research Program study and contributed research materials incorporated into two published department-led books: *Low-Carbon Ecological Space: Rethinking Cross-Dimensional Planning Methods* and *Study on the Economics of Green Buildings in China*.

Assistant Planner, Part-Time, Tongji University Urban Planning and Design Institute 2007–2009

- Supported regional planning, comprehensive urban development planning, tourism planning, historic-district design, and industrial-park planning projects.

Certifications and Professional Credentials

LEED AP Neighborhood Development, U.S. Green Building Council

Honors and Awards

- Dean's Honor List, Albers School of Business and Economics, Seattle University, Fall 2018.
- Best Graduate Design Award, Hebei University of Technology, 2008.
- Second Prize, National Undergraduate Urban Design Competition, Urban Planning Discipline Steering Committee, Ministry of Construction, China, 2007.